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Assessment of jumbo squid in Southeast Pacific Ocean based on state-space biomass dynamics model and impacts of ENSO

China

Assessment of jumbo squid in Southeast Pacific Ocean based on state-space biomass dynamics model and impacts of ENSO

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Executive summary

To account for regional variations in fishing operations, this study incorporated the CPUE CPUE data from Peruvian and Chilean fisheries, and gamma-distribution-standardized CPUE data from the Chinese squid-jigging fishery to develop the relative abundance index for *Dosidicus gigas*. The Bayesian state-space surplus production model revealed sustainable stock status since 2013, with neither overfished status nor overfishing occurring, and annual catch consistently below maximum sustainable yield thresholds.

Among seven environmental-dependent surplus production models evaluated, the Kr model demonstrated superior performance, highlighting the sensitivity to environment variability, particularly through climate-driven fluctuations in carrying capacity and intrinsic growth rate. Notably, despite a significant decline in *D. gigas* landings in 2024, the stock remained healthy—a phenomenon that may be related to the impact of El Niño on population dynamics.

1. Introduction

The jumbo flying squid, *Dosidicus gigas* is a highly migratory cephalopod with a broad distribution across the eastern Pacific Ocean, from California to southern Chile (Nigmatullin et al., 2001). As a key fishery resource supporting coastal economies in Peru, Chile and Ecuador, it also attracts extensive fishing effort from Asia-Pacific nations (Morales-Bojórquez & Pacheco-Bedoya, 2016). Characterized by semelparous reproduction, rapid growth, and high environmental sensitivity, its populations exhibit strong interannual variability linked to ENSO-driven SST and chlorophyll-a fluctuations (Arkhipkin et al., 2021; Robinson et al., 2013). Genetic studies further reveal that *D. gigas* should be managed as a single fishery stock unit throughout the southeastern Pacific (Pardo-Gandarillas et al., 2025). Given its biological plasticity and fishery importance, this study integrated Bayesian state-space surplus production model with environmental dependent factors to assess population dynamics of *D. gigas* in the Southeast Pacific Ocean.

2. Data

2.1 Catch and effort data

Total annual catch data of jumbo flying squid in the Southeast Pacific Ocean during 2012-2023 were derived from Food and Agriculture Organization (FAO) of United Nation (UN) database (www.fao.org/fishery/statistics/global-capture-production/query/en; accessed on 06/17/2025), while data for 2024 was sourced from national fishery statistics.

2.2 Abundance index data

(1) Peruvian fleets data

The abundance index (CPUE) defined as fishing yield per ship was directly from CALAMASUR.

(2) Chilean fleets data

In the Chilean EEZ, jumbo flying squid were caught as by-catch on multispecies trawlers and by direct small-scale hand-jig fishing. The standardized CPUE of trawl and jig fishery were both from CALAMASUR.

(3) Asian fleets data

The CPUE of the Asian fleet was represented by Chinese squid fishing fleet. The data were from China Distant Water Fisheries Association, with field for fishing time (date), fishing area (0.25° longitude ×0.25° latitude) and yield. Nominal CPUE was calculated by dividing catch by fishing days. Generalized Additive Model (GAM) is a common model for standardizing CPUE. The expression of GAM is as follows:

$$g(\mu_i) = \alpha + \sum f_i \times X_i + \varepsilon \tag{1}$$

α is the intercept. f_i is the related coefficient. ε is the error term which is assumed to follow statistical distribution. X_i is an independent variable including environmental factors (i.e. sea surface temperature (SST), concentration of chlorophyll-a (Chl_a), sea surface salinity (SSS) and Niño 1+2 index), spatial-temporal factors (i.e. year, month, longitude, latitude) and interaction terms. Variance inflation factor (VIF<10) indicated no serious multicollinearity between explanatory variables (Table 1).

The assumptions of GAM was that CPUE follows the gamma distribution. For ease of modeling, the 0 value of nominal CPUE was removed.

Table 1. Variance inflation factor (VIF) among explanatory variables

Year	Month	Niño 1+2 index	Latitude	Longitude	SST	SSS	Chl_a
1.140	2.054	1.235	5.113	4.512	4.188	1.799	1.022

2.3 Environmental data

Environmental data came from remote sensing satellite dataset in National Oceanic and Atmospheric Administration (NOAA) database (https://oceanwatch.pifsc.noaa.gov/doc.html, https://origin.cpc.ncep.noaa.gov/data/indices/; accessed on 06/17/2025).

3. Model development

3.1 Description of Bayesian state-space surplus production model

A complete operational surplus production model consists of two parts, i.e. dynamic model which describe the recruitment, growth and mortality of the stock and observation model which relates observation (e.g. CPUE or abundance index by scientific survey) to the stock biomass (Hilborn et al., 1992).

Pella and Tomlinson considered an extension of Schaefer model:

$$B_{y+1} = (B_y + \frac{r}{s-1} B_y (1 - (\frac{B_y}{K})^{s-1} - C_{y,i}) e^{\mu_y} \tag{2}$$

where B_y is the biomass at the beginning of year y; C_y is catch in year y of i; μ_y is process error which follows normal distribution (e.g. μ_y~normal$(0,\tau^2)$), r is the intrinsic growth rate and K is the capacity denoting the unexploited biomass size.

The form of observation model is as follows:

$$I_{y,i} = q_i B_y e^{\varepsilon_y} \tag{3}$$

where i is different fleets, in order of Peru, Asia, Chile_jig and Chile_trawl; $I_{y,i}$ is the index of abundance for year y, fleet i; q_i is catchability coefficient of fleet i ; ε_y is the observation error which follows normal distribution (e.g. ε_y~normal$(0,\sigma^2)$).

3.2 An environmental-dependent surplus production model

Three categories for key parameters (i.e., *r*, *K* and *s*) were determined by sea surface temperature anomaly (SSTA) in the Niño 1+2 area. Parameters *r*_El_Niño, *K*_El_Niño and *s*_El_Niño were used to denote *r*, *K* and *s* for certain years (months) when the SSTA was positive. Parameters *r*_La_Niña, *K*_La_Niña and *s*_La_Niña were used for the rest of the years (months) when the SSTA was negative (Figure 1). Seven models with different structures are as follow:

(1) K model: K1≠K2， r1=r2， s1=s2；
(2) r model: K1=K2， r1≠r2， s1=s2；
(3) s model: K1=K2， r1=r2， s1≠s2；
(4) Kr model: K1≠K2， r1≠r2， s1=s2；
(5) Ks model: K1≠K2， r1=r2， s1≠s2；

(6) rs model: K1=K2，r1≠r2，s1≠s2；
(7) Krs model: K1≠K2，r1≠r2，s1≠s2；

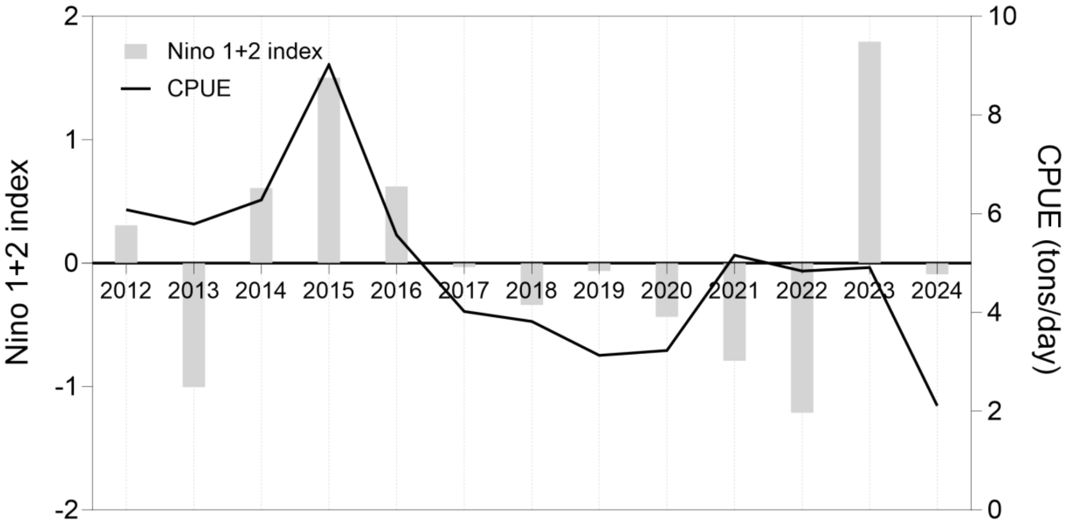


Figure 1 The annual SSTA of Niño 1+2 area and CPUE from year 2012 to 2024

3.3 Prior distribution

Setting appropriate prior distribution is primary in Bayesian paradigm. One base scenario and three sensitivity analysis scenarios were set based on varied prior distribution for traditional model and environmental dependent model. In all scenarios, the standard deviations of process and observation error follow inverse gamma distribution and the rest follow uniform distribution (Table 2 and 3).

Table 2. The prior distributions of parameters for traditional surplus production model

Parameter	Basic Scenario	Sensitivity scenario 1	Sensitivity scenario 2	Sensitivity scenario 3
r	uniform (0.01,5)	uniform (0.4,2)	uniform (0.01,5)	uniform (0.01,5)
K	uniform (1250,25000)	uniform (1250,25000)	uniform (1250,9500)	uniform (1250,25000)
q_1, q_2, q_3, q_4	uniform (0.0001,1)	uniform (0.0001,1)	uniform (0.0001,1)	uniform (0.0001,1)
s	uniform (0.01,5)	uniform (0.1,5)	uniform (0.1,5)	uniform (0.4,2)
B_{2012}/K	uniform (0.01,1)	uniform (0.01,1)	uniform (0.01,1)	uniform (0.01,1)
τ^2	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)
σ^2	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)

Table 3. The prior distributions of parameters for environmental-dependent model

Parameter	Basic Scenario	Sensitivity scenario 1	Sensitivity scenario 2	Sensitivity scenario 3
r_El_Niño r_La_Niña	uniform (0.01,5)	uniform (0.4,2)	uniform (0.01,5)	uniform (0.01,5)
K_El_Niño K_La_Niña	uniform (1250,25000)	uniform (1250,25000)	uniform (1250,9500)	uniform (1250,25000)
q_1, q_2, q_3, q_4	uniform (0.0001,1)	uniform (0.0001,1)	uniform (0.0001,1)	uniform (0.0001,1)
s_El_Niño s_La_Niña	uniform (0.01,5)	uniform (0.01,5)	uniform (0.01,5)	uniform (0.4,2)
B_{2012}/K	uniform (0.01,1)	uniform (0.01,1)	uniform (0.01,1)	uniform (0.01,1)
τ^2	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)
σ^2	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)	inverse gamma (0.001,0.001)

3.4 Posterior distribution and convergence test

Monte Carlo Markov Chain (MCMC) is used to estimate model parameters and biological reference points. The estimation process is carried out in Winbugs. Four chains are used and each chain calculates a total of 50,000 times. The first 10,000 times were discarded and the rest values are reserved once every 10 times. The initial value of each chain is shown in Table 4 and 5.

Convergence of the MCMC samples to the posterior distribution was checked by monitoring the trace of three chains of each parameter. Gelman and Rubin (1992) diagnostics was also examined.

Table 4. The initial value of different MCMC chains in traditional model

Parameter	K	r	q_1, q_2, q_3, q_4	B_{2012}/K	s	τ^2	σ^2
Chain 1	3000	0.9	0.02	0.5	0.9	0.1	0.1
Chain 2	5000	1.1	0.03	0.6	1.1	0.1	0.1
Chain 3	7000	1.3	0.04	0.8	1.3	0.1	0.1
Chain 4	9000	1.5	0.05	1.0	1.5	0.1	0.1

Table 5. The initial value of different MCMC chains in environmental dependent model

Parameter	K_El_Niño	K_La_Niña	r_El_Niño	r_La_Niña	q_1, q_2, q_3, q_4	B_{2012}/K	s	τ^2	σ^2
Chain 1	3000	3000	0.9	0.9	0.02	0.5	0.9	0.1	0.1
Chain 2	5000	5000	1.1	1.1	0.03	0.6	1.1	0.1	0.1
Chain 3	7000	7000	1.3	1.3	0.04	0.8	1.3	0.1	0.1
Chain 4	9000	9000	1.5	1.5	0.05	1.0	1.5	0.1	0.1

3.5 Biological reference points estimation

The biological reference points are essential for figuring out the stock status and giving management advises. The biological references were calculated as follows:

$$B_{MSY} = Ks^{-1/(s-1)} \tag{4}$$

$$F_{MSY} = r/s \tag{5}$$

$$MSY = B_{MSY} \times F_{MSY} \tag{6}$$

4. Results

4.1 Generalized Additive Model

GAM analysis revealed that all the effects on CPUE were significant (P<0.01). The above variables were added to the model one by one and the optimal GAM which had a minimum AIC of 225055.1 incorporates all the significant variables (Table 6). Analysis of variance CPUE showed that year, month, and SSS explained 12.4%, 6.8%, and 3.2%, respectively. The explained deviance of other factors was less than 1% (Table 7). The year effect was used as an abundance index in the stock assessment model. Nominal CPUE and standardized CPUE showed fluctuating trends, apparently opposite in 2012-2014 and 2021-2023, but both were highest in 2015 and lowest in 2024 (Figure 2).

Table 6. Results of GAM selection

GAM	AIC	Deviance explained	R^2
CPUE~Year	234116.9	12.4%	0.118
CPUE~Year+Month	229344.5	19.2%	0.159
CPUE~Year+Month+s(Niño)	228946.9	19.8%	0.165
CPUE~Year+Month+s(Niño)+s(SSS)	226721.0	23.0%	0.193
CPUE~Year+Month+s(Niño)+s(SSS)+s(Chl_a)	226582.3	23.1%	0.193
CPUE~Year+Month+s(Niño)+s(SSS)+s(Chl_a)+s(SST)	226183.2	23.6%	0.196
CPUE~Year+Month+s(Niño)+s(SSS)+s(Chl_a)+s(SST)+s(Latitude)	225695.0	24.3%	0.205
CPUE~Year+Month+s(Niño)+s(SSS)+s(Chl_a)+s(SLA)+s(SST)+ s(Latitude)+s(Longitude)	225396.7	24.6%	0.210
CPUE~Year+Month+s(Niño)+s(SSS)+s(Chl_a)+s(SLA)+s(SST)+ s(Latitude)+s(Longitude)++s(Latitude, Longitude)	225055.1	25.1%	0.212

Table 7. Statistics results of the best GAM

	Year	Month	Niño 1+2 index	SSS	Chl_a	SST	Latitude	Longitude	Latitude ×Longitude
P	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***	<0.01 ***
df	12	11	8.606	8.936	8.337	7.997	7.488	8.465	26.751
Deviance explained	12.4%	6.8%	0.6%	3.2%	0.1%	0.5%	0.7%	0.3%	0.5%

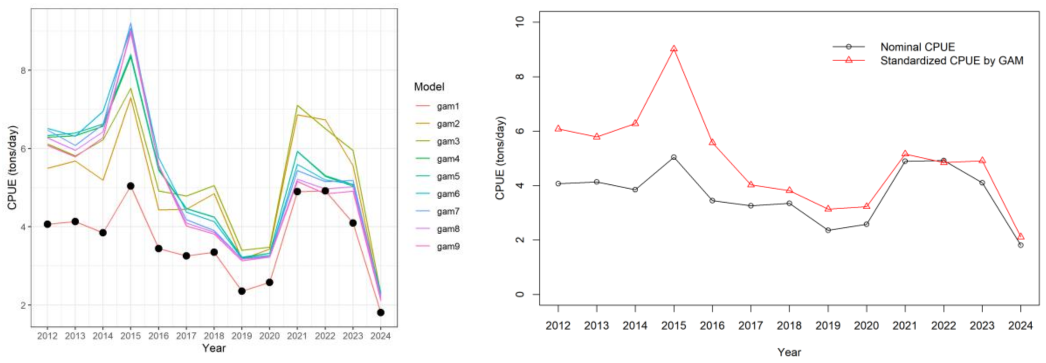


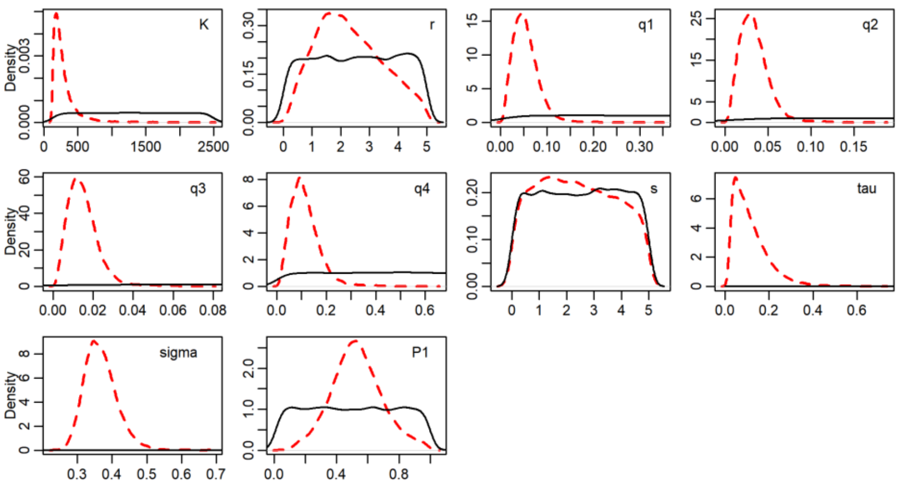
Figure 2. Nominal and standardized CPUE for jumbo flying squid

4.2 Estimates of parameters

The significant difference between prior distribution and posterior distribution indicated that the data in the model provided sufficient information to dominate the form of posterior distribution in all scenarios (Figure 3 and 4).

The estimates of parameters for traditional surplus production model were shown in Table 8. For the base scenario, the mean and the median value of parameters K, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 2.95 and 2.37 million tons, 2.32 and 2.19, 0.054 and 0.051, 0.034 and 0.031, 0.015 and 0.014, 0.113 and 0.104, 2.39 and 2.33, 0.020 and 0.009, 0.137 and 0.131, 0.53 and 0.52, respectively. For the sensitivity analysis scenario 1, the mean and the median value of parameters K, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 4.00 and 3.22 million tons, 1.31 and 1.36, 0.046 and 0.040, 0.029 and 0.025, 0.013 and 0.011, 0.096 and 0.084, 1.57 and 1.28, 0.021 and 0.010, 0.135 and 0.130, 0.50 and 0.49, respectively. For the sensitivity analysis scenario 2, the mean and the median value of parameters K, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 2.73 and 2.31 million tons, 2.38 and 2.28, 0.050 and 0.050, 0.030 and 0.030, 0.010 and 0.010, 0.110 and 0.110, 2.43 and 2.38, 0.019 and 0.009, 0.137 and 0.132, 0.54 and 0.53, respectively. For the sensitivity analysis scenario 3, the mean and the median value of parameters K, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 3.00 and 2.41 million tons, 1.62 and 1.58, 0.058 and 0.053, 0.036 and 0.033, 0.016 and 0.015, 0.121 and 0.111, 1.21 and 1.21, 0.022 and 0.011, 0.138 and 0.133, 0.49 and 0.47, respectively.

According to results of the test, K and Ks model were non-convergent and Kr model with the minimum DIC was the best environmental dependent model (Table 9). The estimates of parameters for Kr model were shown in Table 10. For the base scenario, the mean and the median value of parameters K_El_Niño, K_La_Niña, r_El_Niño, r_La_Niña, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 4.59 and 3.46 million tons, 3.67 and 2.70 million tons, 1.94 and 1.78, 2.29 and 2.17, 0.041 and 0.038, 0.025 and 0.024, 0.011 and 0.010, 0.085 and 0.078, 2.58 and 2.61, 0.014 and 0.006, 0.131 and 0.126, 0.51 and 0.51, respectively. For the sensitivity analysis scenario 1, the mean and the median value of parameters K_El_Niño, K_La_Niña, r_El_Niño, r_La_Niña, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 6.04 and 4.78 million tons, 4.88 and 3.80 million tons, 1.12 and 1.08, 1.36 and 1.39, 0.034 and 0.028, 0.021 and 0.018, 0.009 and 0.008, 0.070 and 0.059, 1.82 and 1.52, 0.013 and 0.006, 0.129 and 0.124, 0.49 and 0.49, respectively. For the sensitivity analysis scenario 2, the mean and the median value of parameters K_El_Niño, K_La_Niña, r_El_Niño, r_La_Niña, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 3.72 and 3.26 million tons, 3.00 and 2.62 million tons, 2.09 and 1.94, 2.40 and 2.29, 0.042 and 0.039, 0.026 and 0.024, 0.011 and 0.010, 0.087 and 0.079, 2.67 and 2.71, 0.012 and 0.006, 0.130 and 0.125, 0.53 and 0.52, respectively. For the sensitivity analysis scenario 3, the mean and the median value of parameters K_El_Niño, K_La_Niña, r_El_Niño, r_La_Niña, q_1, q_2, q_3, q_4, s, σ^2, τ^2 and B_{2012}/K were 4.52 and 3.58 million tons, 3.48 and 2.70 million tons, 1.29 and 1.14, 1.69 and 1.64, 0.042 and 0.039, 0.026 and 0.025, 0.011 and 0.011, 0.088 and 0.081, 1.24 and 1.26, 0.013 and 0.006, 0.130 and 0.126, 0.47 and 0.45, respectively.



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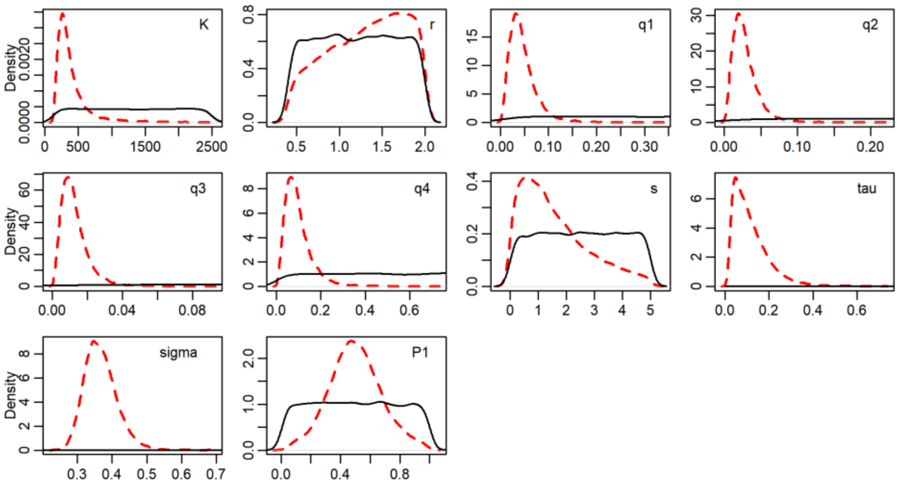

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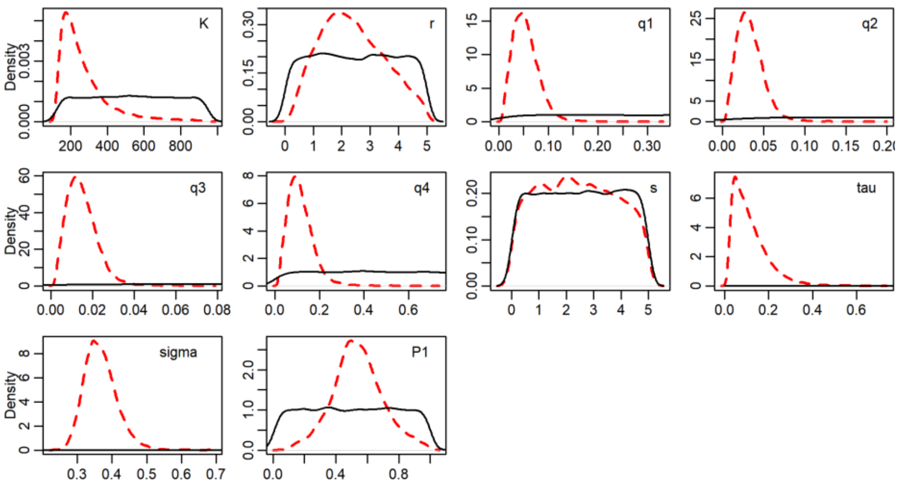

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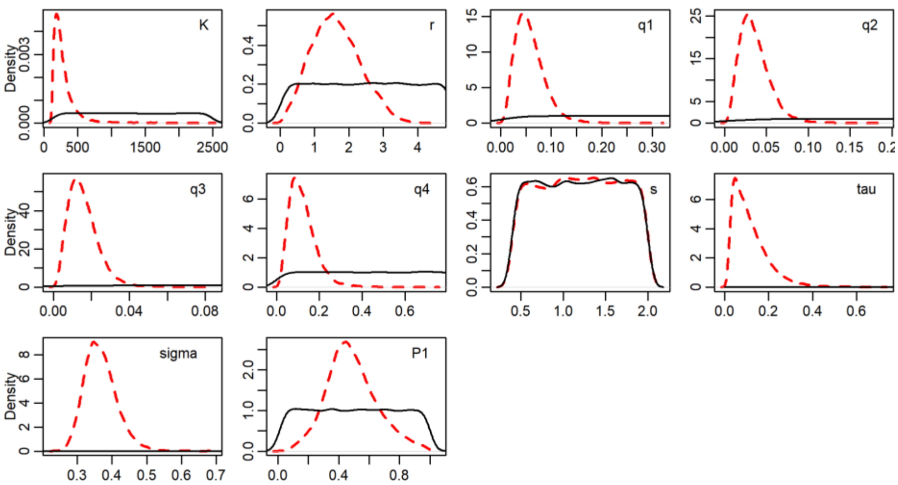
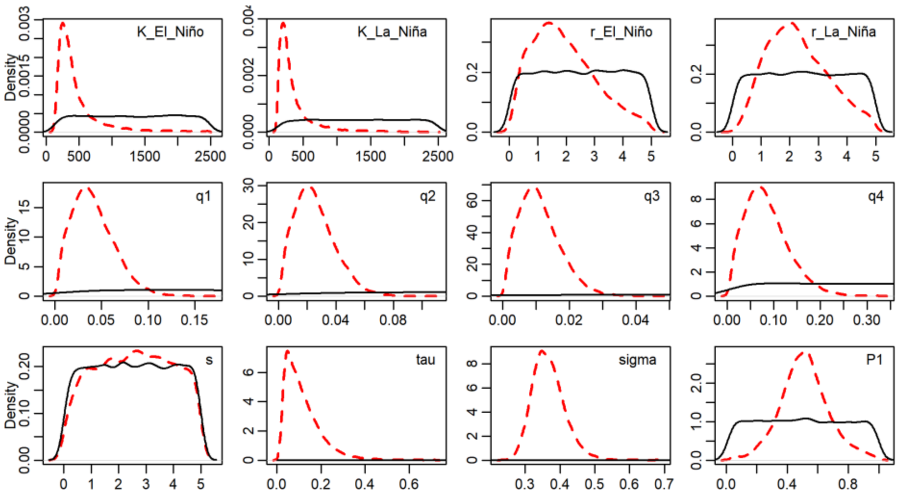


(d)

Figure 3. Prior and posterior distributions of parameters for traditional model (a) base scenario; (b) sensitivity analysis scenario; the unit of K is 10^4 metric ton.

(The figure panels from left to right and from up to bottom respectively represent

K, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2, and P_1)


K_El_Niño
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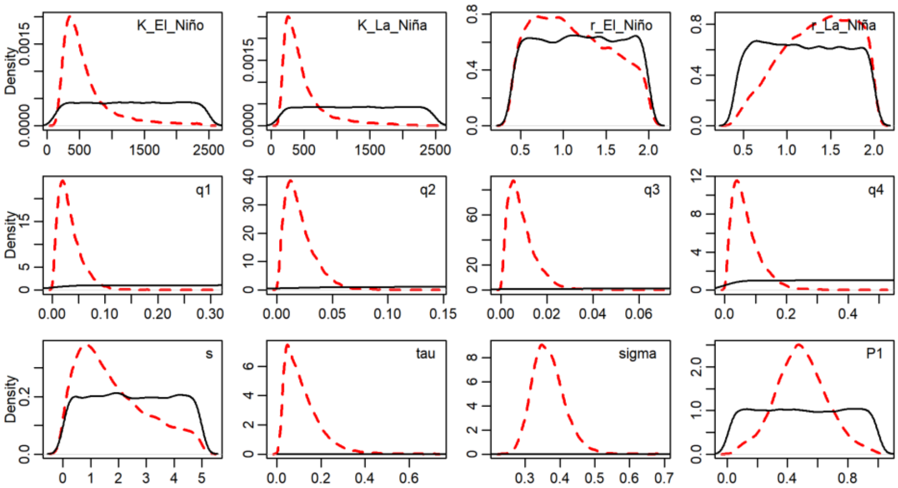
P1
Density

(a)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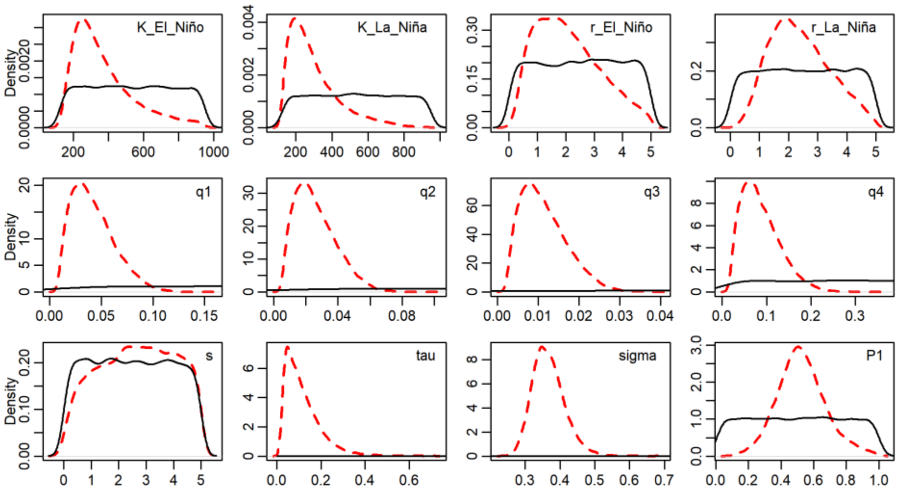
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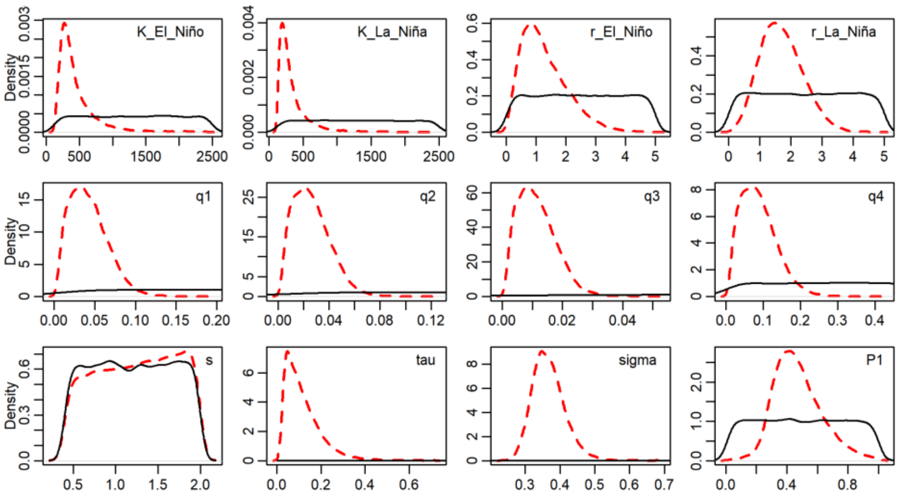

(d)

Figure 4. Prior and posterior distributions of parameters for environmental dependent model (a) base scenario; (b) sensitivity analysis scenario; the unit of K is 10^4 metric ton.

(The figure panels from left to right and from up to bottom respectively represent $K\_El\_Niño$, $K\_La\_Niña$, r, q_1, q_2, q_3, q_4, s, σ^2, τ^2, and P_1)

Table 8 The estimates of parameters for traditional model

Scenario	Statistic	K (kt)	r	q_1	q_2	q_3	q_4	s	σ^2	τ^2	B_{2012}/K
Base scenario	mean	2951.76	2.32	0.054	0.034	0.015	0.113	2.39	0.020	0.137	0.53
	2.50%	1329.00	0.47	0.014	0.009	0.004	0.029	0.15	0.001	0.084	0.22
	25%	1811.00	1.44	0.035	0.022	0.009	0.072	1.22	0.003	0.112	0.42
	50%	2368.00	2.19	0.051	0.031	0.014	0.104	2.33	0.009	0.131	0.52
	75%	3303.00	3.13	0.069	0.043	0.019	0.143	3.54	0.024	0.156	0.63
	97.5%	8210.05	4.66	0.118	0.073	0.032	0.253	4.83	0.100	0.219	0.88
Sensitivity scenario 1	mean	4003.93	1.31	0.046	0.029	0.013	0.096	1.57	0.021	0.135	0.50
	2.50%	1631.97	0.48	0.010	0.006	0.003	0.021	0.08	0.001	0.084	0.16
	25%	2470.00	0.97	0.027	0.017	0.007	0.055	0.65	0.003	0.111	0.38
	50%	3218.00	1.36	0.040	0.025	0.011	0.084	1.28	0.010	0.130	0.49
	75%	4491.25	1.69	0.058	0.036	0.016	0.121	2.24	0.025	0.153	0.61
	97.5%	12000.25	1.97	0.118	0.074	0.032	0.248	4.43	0.102	0.217	0.87
Sensitivity scenario 2	mean	2728.59	2.38	0.050	0.030	0.010	0.110	2.43	0.019	0.137	0.54
	2.50%	1342.00	0.55	0.020	0.010	0.000	0.030	0.16	0.001	0.084	0.22
	25%	1788.00	1.53	0.040	0.020	0.010	0.070	1.26	0.003	0.112	0.44
	50%	2306.50	2.28	0.050	0.030	0.010	0.110	2.38	0.009	0.132	0.53
	75%	3179.00	3.19	0.070	0.040	0.020	0.150	3.56	0.023	0.156	0.64
	97.5%	6842.02	4.63	0.110	0.070	0.030	0.250	4.82	0.096	0.222	0.88
Sensitivity scenario 3	mean	3001.19	1.62	0.058	0.036	0.016	0.121	1.21	0.022	0.138	0.49
	2.50%	1342.00	0.40	0.016	0.010	0.004	0.033	0.44	0.001	0.083	0.19
	25%	1862.00	1.11	0.037	0.023	0.010	0.076	0.81	0.004	0.112	0.38

50%	2413.00	1.58	0.053	0.033	0.015	0.111	1.21	0.011	0.133	0.47
75%	3327.25	2.10	0.074	0.046	0.020	0.153	1.60	0.027	0.158	0.58
97.5%	8444.27	3.08	0.126	0.078	0.035	0.267	1.96	0.107	0.222	0.85

Table 9 Results of statistics fitted to environmental dependent models

Model	K model	r model	s model	Kr model	Ks model	rs model	Krs model
DIC	59.429	175.837	172.909	130.909	99.254	171.121	137.683

Table 10 The estimates of parameters for environmental dependent model

Scenario	Statistic	K_El_Niño	K_La_Niña	r_El_Niño	r_La_Niña	q_1	q_2	q_3	q_4	s	σ^2	τ^2	B_{2012}/K
Base scenario	mean	4587.48	3674.61	1.94	2.29	0.041	0.025	0.011	0.085	2.58	0.014	0.131	0.51
	2.50%	1641.00	1370.00	0.30	0.57	0.007	0.005	0.002	0.015	0.22	0.001	0.081	0.20
	25%	2518.00	2010.75	1.08	1.48	0.024	0.015	0.007	0.050	1.46	0.002	0.108	0.41
	50%	3455.00	2701.00	1.78	2.17	0.038	0.024	0.010	0.078	2.61	0.006	0.126	0.51
	75%	5177.25	4089.00	2.66	3.01	0.055	0.034	0.015	0.112	3.72	0.015	0.148	0.61
	97.5%	15600.25	12741.00	4.44	4.58	0.092	0.057	0.025	0.194	4.86	0.075	0.210	0.86
Sensitivity	mean	6040.70	4882.28	1.12	1.36	0.034	0.021	0.009	0.070	1.82	0.013	0.129	0.49
scenario 1	2.50%	2090.95	1698.97	0.44	0.57	0.006	0.004	0.002	0.013	0.11	0.001	0.081	0.15
	25%	3454.75	2685.00	0.75	1.06	0.018	0.011	0.005	0.036	0.80	0.002	0.106	0.38
	50%	4775.00	3796.00	1.08	1.39	0.028	0.018	0.008	0.059	1.52	0.006	0.124	0.49
	75%	7054.00	5720.50	1.48	1.69	0.045	0.028	0.012	0.092	2.64	0.015	0.146	0.60
	97.5%	18570.50	15410.50	1.94	1.97	0.089	0.055	0.024	0.184	4.65	0.069	0.206	0.85
Sensitivity	mean	3722.11	3004.68	2.09	2.40	0.042	0.026	0.011	0.087	2.67	0.012	0.130	0.53
scenario 2	2.50%	1599.00	1370.97	0.42	0.73	0.013	0.008	0.003	0.026	0.28	0.001	0.081	0.25
	25%	2451.00	1969.00	1.19	1.63	0.026	0.016	0.007	0.053	1.57	0.002	0.107	0.43
	50%	3264.50	2622.50	1.94	2.29	0.039	0.024	0.010	0.079	2.71	0.006	0.125	0.52
	75%	4560.25	3652.25	2.85	3.09	0.055	0.034	0.015	0.112	3.81	0.014	0.148	0.61
	97.5%	8210.00	6764.05	4.54	4.53	0.089	0.055	0.024	0.188	4.88	0.064	0.205	0.85

Sensitivity scenario 3	mean	4519.45	3478.18	1.29	1.69	0.042	0.026	0.011	0.088	1.24	0.013	0.130	0.47
	2.50%	1669.97	1388.97	0.22	0.52	0.009	0.005	0.002	0.017	0.45	0.001	0.081	0.22
	25%	2655.00	2001.00	0.71	1.20	0.025	0.015	0.007	0.050	0.85	0.002	0.107	0.36
	50%	3580.50	2698.00	1.14	1.64	0.039	0.025	0.011	0.081	1.26	0.006	0.126	0.45
	75%	5294.00	4006.25	1.74	2.14	0.057	0.035	0.015	0.117	1.64	0.014	0.148	0.56
	97.5%	13300.50	10291.25	3.11	3.14	0.093	0.058	0.025	0.198	1.97	0.070	0.208	0.82

4.3 Biological reference points

Three biological reference points were estimated. The results from traditional model were shown in Table 11. For the base scenario, the mean and median value of MSY were 1.35 and 1.19 million tons. The mean and median value of F_{MSY} were 1.71 and 0.05. The mean and median value of B_{MSY} were 1.43 and 1.14 million tons. For the sensitivity analysis scenario 1, the mean and median value of MSY were 1.50 and 1.28 million tons. The mean and median value of F_{MSY} were 2.17 and 1.01. The mean and median value of B_{MSY} were 1.75 and 1.26 million tons. For the sensitivity analysis scenario 2, the mean and median value of MSY were 1.33 and 1.19 million tons. The mean and median value of F_{MSY} were 1.64 and 1.05. The mean and median value of B_{MSY} were 1.33 and 1.13 million tons. For the sensitivity analysis scenario 3, the mean and median value of MSY were 1.36 and 1.22 million tons. The mean and median value of F_{MSY} were 1.48 and 1.37. The mean and median value of B_{MSY} were 1.16 and 0.93 million tons.

The results from environmental dependent model were shown in Table 12. For the base scenario, the mean and median value of MSY_El_Niño and MSY_La_Niña were 1.64 and 1.27 million tons, 1.60 and 1.30 million tons, respectively. The mean and median value of F_{MSY}_El_Niño and F_{MSY}_La_Niña were 1.26 and 0.77, 1.48 and 0.95, respectively. The mean and median value of B_{MSY}_El_Niño and B_{MSY}_La_Niña were2.34 and 1.71 million tons, 1.89 and 1.38 million tons, respectively. For the sensitivity analysis scenario 1, the mean and median value of MSY_El_Niño and MSY_La_Niña were 1.85 and 1.40 million tons, 1.77 and 1.43 million tons, respectively. The mean and median value of F_{MSY}_El_Niño and F_{MSY}_La_Niña were 1.63 and 0.69, 1.84 and 0.90, respectively. The mean and median value of B_{MSY}_El_Niño and B_{MSY}_La_Niña were 2.85 and 2.06 million tons, 2.31 and 1.63 million tons, respectively. For the sensitivity analysis scenario 2, the mean and median value of MSY_El_Niño and MSY_La_Niña were 1.50 and 1.26 million tons, 1.43 and 1.26 million tons, respectively. The mean and median value of F_{MSY}_El_Niño and F_{MSY}_La_Niña were 1.23 and 0.80, 1.43 and 0.95, respectively. The mean and median value of B_{MSY}_El_Niño and B_{MSY}_La_Niña were 1.92 and 1.65 million tons, 1.56 and 1.35 million tons, respectively. For the sensitivity analysis scenario 3, the mean and median value of MSY_El_Niño and MSY_La_Niña were 1.64 and 1.25 million tons, 1.70 and 1.41 million tons, respectively. The mean and median value of F_{MSY}_El_Niño and F_{MSY}_La_Niña were 1.16 and 0.96, 1.51 and 1.40, respectively. The mean and median value of B_{MSY}_El_Niño and B_{MSY}_La_Niña were 1.78 and 1.39 million tons, 1.37 and 1.06 million tons, respectively.

Table 11 The estimates of biological reference points for traditional model

Scenario	Statistic	MSY(kt)	F_{MSY}	B_{MSY}(kt)
	mean	1345.63	1.71	1429.94
	2.50%	923.90	0.33	279.40
Base scenario	25%	1060.00	0.76	829.30
	50%	1189.00	1.05	1140.00
	75%	1438.00	1.46	1636.25

	97.5%	2734.02	6.52	4397.15
	mean	1503.27	2.17	1747.76
	2.50%	906.99	0.26	164.39
Sensitivity	25%	1077.00	0.61	744.27
scenario 1	50%	1275.00	1.01	1258.00
	75%	1640.00	1.83	2124.00
	97.5%	3486.07	12.19	6548.20
	mean	1326.29	1.64	1327.67
	2.50%	924.00	0.39	279.19
Sensitivity	25%	1059.00	0.78	828.97
scenario 2	50%	1186.00	1.05	1130.00
	75%	1418.25	1.45	1600.00
	97.5%	2595.12	6.32	3648.10
	mean	1359.79	1.48	1159.51
	2.50%	923.90	0.39	448.69
Sensitivity	25%	1069.00	0.99	707.37
scenario 3	50%	1220.00	1.37	927.60
	75%	1490.00	1.81	1306.00
	97.5%	2604.00	3.32	3257.02

Table 12 The estimates of biological reference points for environmental dependent model

Scenario	Statistic	EI_Niño			La_Niña		
		MSY	F_{MSY}	B_{MSY}	MSY	F_{MSY}	B_{MSY}
Base scenario	mean	1641.81	1.26	2339.52	1597.85	1.48	1887.49
	2.50%	676.59	0.16	482.28	853.69	0.27	384.39
	25%	1019.00	0.52	1201.00	1089.00	0.66	948.32
	50%	1271.00	0.77	1711.00	1296.00	0.95	1374.00
	75%	1747.00	1.12	2654.00	1678.00	1.38	2132.00
	97.5%	5279.07	4.28	8418.70	4507.10	4.96	6904.05
	mean	1849.35	1.63	2847.13	1769.86	1.84	2310.39
Sensitivity	2.50%	794.49	0.18	286.19	872.20	0.23	241.40
	25%	1082.00	0.41	1156.00	1142.00	0.51	900.62
scenario 1	50%	1399.00	0.69	2060.50	1431.00	0.90	1632.00
	75%	2074.00	1.27	3547.00	1976.00	1.66	2884.75
	97.5%	5576.02	8.37	10570.00	4789.20	9.45	8796.02
	mean	1498.11	1.23	1920.52	1426.55	1.43	1560.99
	2.50%	723.39	0.23	545.28	861.70	0.35	418.79
Sensitivity	25%	1023.00	0.56	1184.75	1077.00	0.68	944.10
scenario 2	50%	1256.00	0.80	1654.00	1259.00	0.95	1350.00
	75%	1681.00	1.12	2403.25	1570.00	1.34	1967.00
	97.5%	3671.05	4.14	4651.05	3049.02	4.33	3842.12
Sensitivity	mean	1639.70	1.16	1781.46	1703.04	1.51	1374.28
scenario 3	2.50%	621.49	0.20	596.40	905.29	0.44	475.30

	25%	985.42	0.62	1018.00	1171.00	1.01	768.90
	50%	1246.00	0.96	1394.50	1413.00	1.40	1055.00
	75%	1785.25	1.47	2084.00	1856.00	1.87	1580.00
	97.5%	5128.02	3.28	5422.07	4380.10	3.31	4267.07

4.5 Stock status

Time-series analysis of Bratio (B/B_{MSY}) and Fratio (F/F_{MSY}) revealed consistent trends under all management scenarios. Kobe plot diagnostics confirmed the fishery has maintained sustainable status since 2013, with no instances of overfishing or overfished. The traditional stock assessment model projected terminal-year stock health probabilities of 58.6, 73.1%, 59.3%, and 72.3% across scenarios (Figure 5). The probability of healthy stock status increased to a range of 70.9%-88.3% in the environment dependent model (Figure 6), demonstrating the critical role of environmental factors on population biomass dynamics.

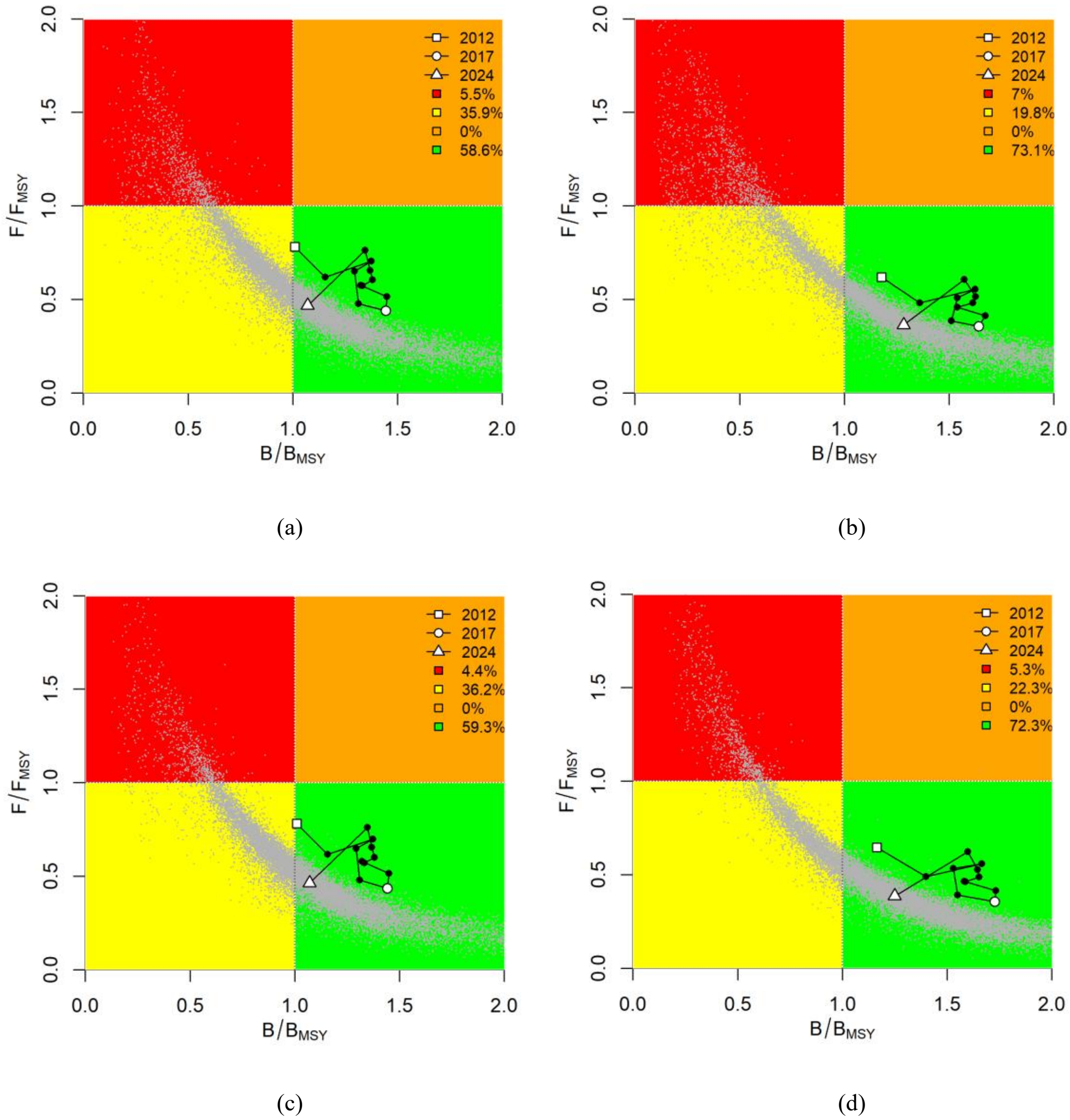

Figure 5. Kobe plot of traditional model, (a) base scenario, (b) sensitivity analysis scenario 1, (c) sensitivity analysis scenario 2, (d) sensitivity analysis scenario 3

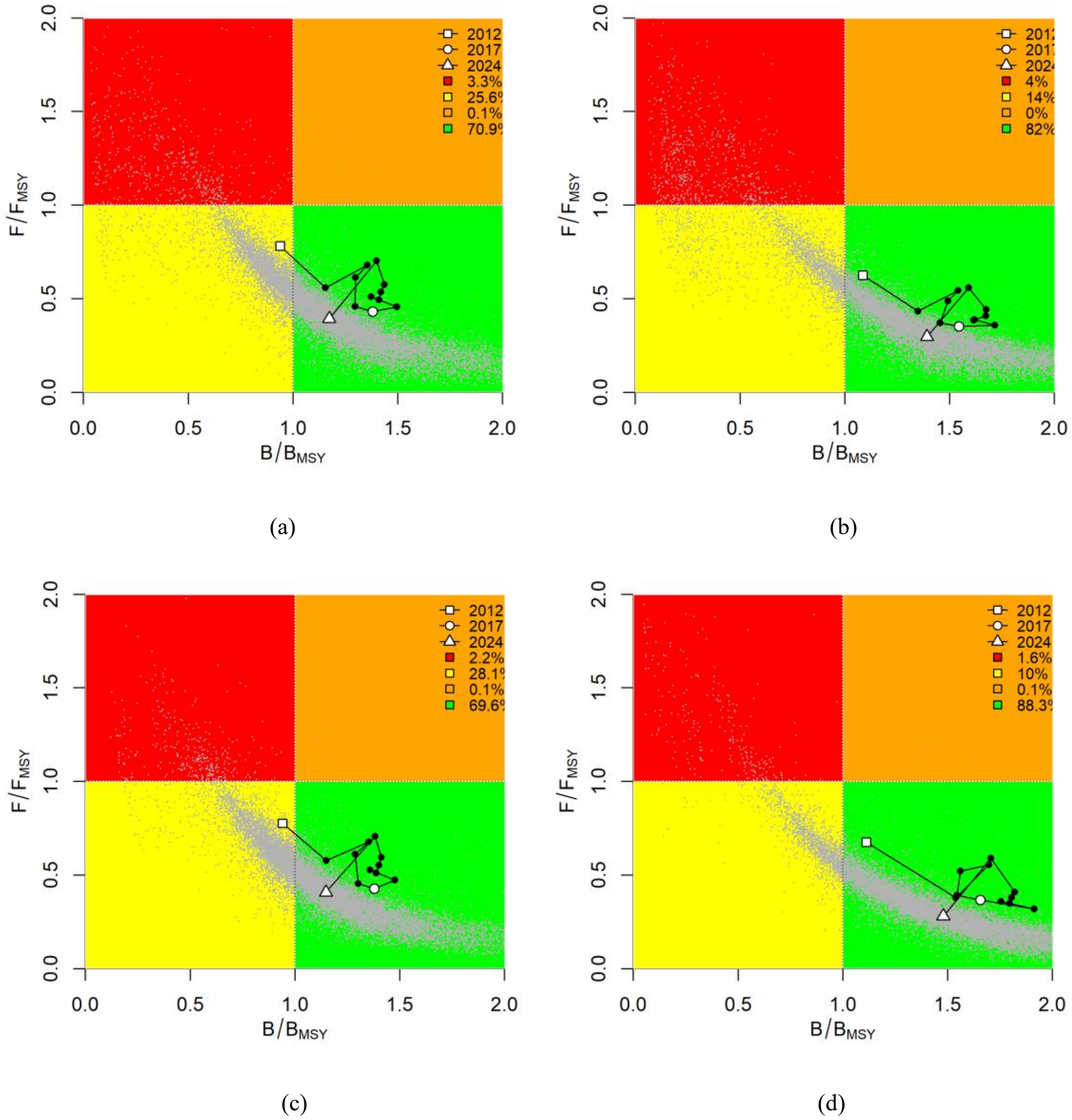

Figure 6. Kobe plot of environmental dependent model, (a) base scenario, (b) sensitivity analysis scenario 1, (c) sensitivity analysis scenario 2, (d) sensitivity analysis scenario 3

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